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Phytochemistry and therapeutic potential of *Bixa orellana* L.

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Abstract
Often referred to as annatto, lipstick tree, or achiote, *Bixa orellana* L. (*Bixa*ceae) is a perennial shrub or small tree that is found throughout tropical and subtropical areas. The carotenoid pigments bixin and norbixin, which are widely used as natural colorants in food, pharmaceutical, cosmetic, and industrial products, are found in the plant's seeds. In addition to its economic significance, B. *orellana* has long been used traditionally in indigenous medicine to treat skin conditions, gastrointestinal issues, infections, and inflammatory diseases. Numerous bioactive components, such as carotenoids, flavonoids, phenolic compounds, terpenoids, tannins, essential oils, and fixed oils, have been identified through phytochemical research. Which together support its many pharmacological properties, including hepatoprotective, immunomodulatory, antioxidant, antimicrobial, anti-inflammatory, antidiabetic, and anticancer effects. B. *orellana* has drawn more attention from scientists and businesses due to its medicinal potential and safety as a natural colorant. The botanical characteristics, phytochemical makeup, therapeutic qualities, and industrial significance of *Bixa orellana* as a useful plant resource for natural additives and plant-based medicines are highlighted in this abstract.

Keywords: *Bixa orellana* L, annatto, achiote, bixin, norbixin, medicinal plant, phytochemistry, ethnomedicine, antioxidant activity, antimicrobial activity, anti-inflammatory activity, pharmacological properties, natural food colorant,industrial applications, traditional medicine

Introduction
Bixa orellana L., also referred to as lipstick tree, annatto, or achiote, is a perennial shrub in the *Bixa*ceae family that grows widely in tropical and subtropical areas of Asia, Africa, and Central and South America. The plant has long been grown for its seeds, which are rich in natural pigments, mainly bixin and norbixin, which are widely used as food coloring in the dairy, cosmetic, and pharmaceutical industries [1]. Beyond its commercial value, indigenous populations have long used B. *orellana* ethnomedically to treat a variety of illnesses, such as microbial diseases, skin infections, gastrointestinal issues, and inflammation [2].
According to phytochemical studies, B. *orellana* is a storehouse of bioactive substances that contribute to its various pharmacological activities, including carotenoids, flavonoids, terpenoids, tannins, and essential oils. Several plant parts, especially the seeds and leaves, have been shown in experiments to have antioxidant, antimicrobial, anti-inflammatory, anticancer, antidiabetic, and hepatoprotective qualities [3]. B. *orellana* has garnered substantial scientific attention in recent decades due to the growing interest in natural additives and plant-based medicines [4]. In order to highlight *Bixa orellana*'s significance as a useful medicinal and industrial plant, this review attempts to give a thorough overview of the plant's botanical traits, phytochemistry, traditional uses, and pharmacological potential (5).

Botanical Description
Bixa orellana L. is an evergreen shrub or small tree in the *Bixa*ceae family that usually reaches a height of three to ten meters [6]. The plant has a well-branched canopy and a short trunk with soft, light-brown to grayish bark. When young branches reach adulthood, they become glabrous [7].
The simple, alternating, ovate to broadly cordate leaves have an acuminate apex, an entire margin, and a length of about 8 to 20 cm [8]. Their lower surface may have sparse hairs,

but their upper surface is glabrous, thin, and bright green. The leaves have noticeable palmate venation and are supported by long petioles that are typically 5-10 cm long. Terminal or axillary panicles bear large, striking, bisexual flowers. Each flower typically has five pink to white petals and numerous prominent stamens with yellow anthers, adding to its ornamental appeal. Flowering, which usually occurs during warm seasons, is influenced by climate conditions [9]. The fruit is capsular, ovoid to globose, and 3 to 5 cm in diameter. Its flexible, soft spines set it apart. The capsule dehisces to reveal a bright red or orange pulp with numerous angular seeds when it reaches maturity. The plant's seeds, which are rich in carotenoid pigments, especially norbixin and bixin, are its primary commercial product [10]. *Bixa orellana* thrives in tropical and subtropical climates and prefers full sun and well-drained soils. It is widely grown in parts of Africa,Southeast Asia, the Caribbean, and South and Central America for its industrial, medicinal, and decorative uses [11].

Chemical Constituents

Chemically, *Bixa orellana* L. has a wide variety of bioactive components, the most well-known and thoroughly researched of which are carotenoids [12]. These pigments, which give the plant its distinctive reddish-orange hue and substantial industrial value, are mostly found in the seeds [13].

Carotenoids

Bixin, a lipophilic apocarotenoid that makes up as much as 80% of the total carotenoid content, is the main carotenoid found in B. *orellana* seeds. Saponification produces norbixin, a water-soluble dicarboxylic derivative of bixin that is frequently utilized in pharmaceutical and food formulations. There are also known minor carotenoids, including lutein, zeaxanthin, cryptoxanthin, β-carotene, and isobixin [14]. These substances play a major role in the plant's chemoprotective and antioxidant qualities [15].

Flavonoids and Phenolic Compounds

The leaves, seeds, and bark of B. *orellana* have been found to contain a variety of phenolic acids and flavonoids [16]. These include substances like derivatives of kaempferol, quercetin, catechin, salicylic acid, and ellagic acid. The plant's antimicrobial, anti-inflammatory, and antioxidant properties are linked to the presence of these polyphenolic components [17].

Volatile Constituents and Terpenoids

Several monoterpenes and sesquiterpenes, such as α-pinene, β-pinene, limonene, β-caryophyllene, and spathulenol, are found in the essential oil extracted from the leaves and seeds [18]. The plant's antimicrobial and insect-repelling qualities are thought to be influenced by these volatile components [19].

Additional Constituents

Tannins, saponins, fixed oils, mucilage, sterols, and proteins are among the other phytochemicals found in B. *orellana*. It has been demonstrated that the leaves contain significant amounts of vitamins and minerals, and the seeds contain a fixed oil rich in unsaturated fatty acids [20]. These components work together to support the plant's wide pharmacological potential in both conventional and contemporary medicine [21].

Medicinal Uses

1. Used in traditional medicine to treat pain, swelling, and inflammatory conditions as an analgesic and anti-inflammatory.
2. Decoctions of leaves are used as an antipyretic to treat fever [22].
3. Historically used to treat gastrointestinal issues such as stomach ulcers, diarrhea, and dysentery.
4. Because of its antimicrobial qualities, it is applied topically to wounds, burns, skin conditions, and microbial infections.
5. As an antioxidant, it aids in lowering lipid peroxidation and oxidative stress [23].
6. Its use in metabolic disorders is supported by reports that it has antihyperlipidemic and antidiabetic properties.
7. Has hepatoprotective properties that support detoxification and liver protection.
8. Used to treat intestinal worm infestations as an anthelmintic [24].
9. Used in conventional methods to treat respiratory conditions like asthma and bronchitis.
10. Experimental research revealed immunomodulatory and anticancer [25].

Industrial Uses

1. The main natural source of annatto, a common natural food coloring.
2. Bixin and norbixin pigments are used to color beverages, cheese, butter, margarine, dairy products, and baked goods.
3. Used as a coloring agent in tablets, capsules, and ointments, among other pharmaceutical formulations [26].
4. Used in the cosmetics industry for hair products, coloring creams, soaps, and lipsticks.
5. Used as a natural textile dye and paint and leather coloring agent [27].
6. Indigenous communities have historically used it as sunscreen and body paint.
7. Used as a natural insecticide
8. Used as an environmentally friendly coloring agent in the polymer and plastic industries.
9. Because it is safe and biodegradable, it is valued as a natural substitute for synthetic dyes [28].
10. Because consumers prefer natural additives, they are being used more frequently in clean-label and organic products [29].

Conclusion

With a long history of traditional use and well-established commercial significance, *Bixa orellana* L. is a valuable medicinal and industrial plant. Its use in the treatment of inflammatory conditions, infections, gastrointestinal issues, metabolic diseases, and skin conditions is supported by a wealth of ethnobotanical knowledge.

Many of these traditional claims have been confirmed by recent phytochemical and pharmacological studies, which attribute the plant's therapeutic potential to its rich composition of bioactive compounds, including flavonoids,

phenolic acids, volatile constituents, and carotenoids like norbixin and bixin.

Apart from its therapeutic value, B. *orellana* is an important natural colorant in a number of industries, particularly the food, pharmaceutical, and cosmetic industries. The industrial significance of annatto made from B. *orellana* has been further increased by the growing demand for natural, safe, and sustainable additives worldwide. There is still little clinical validation and standardization of extracts, despite strong experimental evidence for their biological activities.

To fully realize the medicinal and commercial potential of this plant, future research should concentrate on carefully planned clinical trials, toxicological analyses, and the creation of standardized formulations. All things considered, *Bixa orellana* is a promising natural resource that connects conventional medicine with contemporary industrial uses, justifying further research.

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